$$\frac{1}{2}mv^2 - G\frac{M_s m}{a} = \frac{1}{2}mv_p^2 - \frac{GmM_s}{\overline{SP}} \ . \qquad (6)$$

Earth's orbital period T is known ($T = 365$ d). For Earth's mass M_e, we have

$$G\frac{M_s M_e}{a^2} = M_e\left(\frac{2\pi}{T}\right)^2 a \ . \qquad (7)$$

Solving (3)-(7) with the data gives

$$v = 29.8 \text{ km/s} \ . \qquad (8)$$

(8) gives v as the craft's solar-frame speed at E, after escaping Earth's gravity. E is the entry point. Let the speed at E relative to Earth be u ,

Earth's orbital velocity about the Sun is

$$v_e = \frac{2\pi a}{T} = 29.8 \text{ km/s} \ , \qquad (9)$$

Its direction is shown in Fig. 1. Velocity addition gives

$$v = u + v_e \ , \qquad (10)$$

The velocity diagram is Fig. 2; v_e is normal to $\overline{ES}$; hence

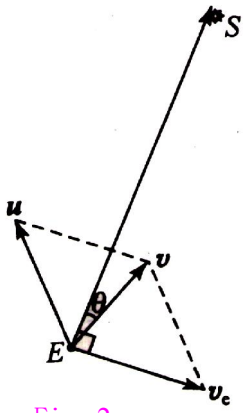


Fig. 2

$$u = \sqrt{v^2 + v_e^2 - 2vv_e\cos\left(\frac{\pi}{2} - \theta\right)} \ , \qquad (11)$$

Substituting the data gives

$$u = 39.1 \text{ km/s} \ . \qquad (12)$$

u is the Earth-relative speed at E, rather than the launch speed u_0. To find u_0 use a frame fixed to Earth's center. At launch the Earth-relative speed is u_0 and the distance is Earth's radius R_e. At Earth-relative speed u, Earth's gravity is negligible. Conservation of energy gives